GuardLink: Dynamic Linking of CVE to MITRE ATT&CK Techniques using Machine Learning

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***Abstract*—As our dependence on digital technologies continues to expand, the need to strengthen our cyber defenses against the increasing threats posed by malicious entities becomes more critical. While existing cybersecurity frameworks and databases like MITRE ATT&CK and Common Vulnerabilities and Exposures (CVE) offer valuable insights for effective threat mitigation, they often operate independently, leading to siloed information. We assert that the automatic linking of vulnerabilities from the CVE database to MITRE ATT&CK adversarial techniques and tactics, particularly focusing on new ones, can provide essential information to empower blue teams in enhancing their cybersecurity defenses against cyberattacks. Gaining proactive insight into an attacker's potential next moves is pivotal for effective defense strategies. Therefore, we introduce in this paper GuardLink, a dynamic approach for linking CVE identifiers (IDs) to MITRE ATT&CK techniques. We first studied, reproduced, evaluated, and improved state of the art models in the field. Furthermore, we proposed a new multi-label classification model that outperforms the existing ones and achieves an accuracy of 97.83%. To ensure transparency and reproducibility, the source code for GuardLink is made openly accessible on GitHub.**

***Index Terms*—CVE, MITRE ATT&CK, Techniques, Tactics, Vulnerability, Machine Learning, Deep Learning, Mapping, Multi-label, Classification**

I. Introduction

In the era of escalating global connectivity, our reliance on digital technology has surged dramatically. It appears that a majority of individuals find themselves immersed in a digital realm, shaping both their personal and professional spheres. This digital revolution, while delivering unparalleled convenience and efficiency, has concurrently unveiled a landscape of challenges in terms of threats and vulnerabilities that demand both academic and industry attention. It is evident from technology advances that the strategies employed by cybercriminals, state-sponsored actors, and other malicious entities evolve in tandem. The continuous progression of technology not only enhances our daily lives but also necessitates a proactive response to address the ever-evolving tactics employed by those who seek to exploit digital vulnerabilities.

At the heart of contemporary cybersecurity lies a fundamental need to comprehensively comprehend and effectively address vulnerabilities. Therefore, we considered in this paper the Common Vulnerabilities and Exposures (CVE) [1] list and the MITRE ATT&CK Framework [2]. These components take an imoprtant role in the battle against cyber threats by offering meticulously structured methodologies for managing vulnerabilities and detecting potential threats. The Common Vulnerabilities and Exposures (CVE) [1] is a global registry encompassing known vulnerabilities and security flaws across software, hardware, and various technological domains. Serving as a standardized and easily comprehensible reference, CVE list provides the cybersecurity community with a common language to identify and describe vulnerabilities formally.

Moreover, the MITRE ATT&CK framework [2] serves as a repository of knowledge encompassing adversary tactics, techniques, and sub-techniques employed to exploit vulnerabilities and inflict impacts. This framework delineates the methods attackers employ to target organizations, offering a valuable guideline for classifying and describing cyber attacks and intrusions. Beyond that, it supplies additional pertinent information for both the detection and mitigation of such methods. The focal point of our paper is the synergistic relationship between CVEs and the MITRE ATT&CK framework [2]. Despite both originating from the MITRE Corporation, these entities exist independently from each other. Previous efforts have sought to establish links between them using machine learning models. This paper objective is to demonstrate how establishing a connection between CVEs and MITRE techniques can significantly enhance the capacity of organizations to safeguarding their assets from cyber attacks. Such a linkage aims to bolster the ability to respond effectively to emerging threats and vulnerabilities, providing the requisite knowledge and tools to stay one step ahead of malicious entities with intentions to target them.

The objective of this paper is to explore and construct a threat intelligence platform designed to predict with a high precision a link that associates MITRE technique ID(s) to a given CVE ID. Achieving this goal involves leveraging machine learning algorithms capable of identifying vulnerabilities and threats that can lead to cybersecurity incidents within a system, and offering effective mitigation solutions in response. Therefore, the main contributions of this paper can be summarized as follows:

1) **Scraping and Fixing Data:** to reproduce the identified

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models, we collected and curated datasets from the state-of-the-art papers. Then, we aligned them with the current MITRE Framework version by updating technique IDs and addressing the deprecated entries.

2) **Model Development:** we propose GuardLink, a novel solution based on Classifier Chain with Logistic Regression, which demonstrated a significant improvement in performance compared to the state-of-the-art models.

Furthermore, we performed a data augmentation and performance testing by using hyperparameter tuning techniques to further improve the designed and implemented solutions to reach an accuracy of 97.83% for our GuardLink model.

In order to contribute further to the research effort in the field, we made the design choice from the early stages of the process to make our work reproducible. Indeed, we publicly share our source code along with the updated and curated dataset on our GitHub repository[1].

The subsequent sections of this paper are structured as follows: In Section II, we provide a large overview of the key works in the field of linking CVE IDs to MITRE ATT&CK technique IDs. In Section III, we describe the theoretical architecture of our proposed solution, GuardLink. The process for handling datasets utilized in our experiments is detailed in Section IV. In Section V, we showcase the novel models examined for linking CVE IDs to techniques IDs, including the selected model for GuardLink and the associated experiments. The paper is brought to a conclusion in Section VI, where we offer insights into potential future research directions.

II. State of the art

In the literature, multiple efforts have been carried out to link the CVE [1] database to the MITRE ATT&CK framework [2]. This section aims to spotlight key papers in this domain, providing an overview of each proposed solution.

In [3], a self-distillation approach was employed by the authors to establish automated links between CVE and MITRE tactics. They introduced a CVET architecture, leveraging a BERT-based model, specifically RoBERTa, fine-tuned with an Adam loss function that included bias correction and a self-KD (Knowledge Distillation) design. The experiment utilized a dataset comprising 24,863 CVEs associated with 10 out of the 14 available MITRE Tactics. The proposed architecture exhibited superior performance compared to various state-of-the-art models across classical, Deep, and pre-Trained Language Models. However, the scope of this experiment focused only on linking CVEs to MITRE tactics, which encompassed only 14 tactics in total. Given that the ultimate goal is to establish links between CVEs and MITRE Techniques, totaling 201, as opposed to the 14 Tactics, the applicability of the CVET architecture may face limitations in this broader context. Grigorescu et al. in [4] sought to establish connections between CVEs and MITRE ATT&CK techniques utilizing BERT-based language models. The model's objective was to perform this linkage based on the information contained within CVE descriptions. Additionally, the authors explored the application of Data Augmentation to investigate its potential to enhance the model and address class imbalance issues. This was particularly relevant as a substantial number of CVEs were associated with a limited set of techniques, while other techniques had minimal or no examples for learning. Among the various classical and Deep Learning approaches employed, the Multi-Label SciBERT with Data Augmentation emerged as the most effective, demonstrating the highest results with a weighted F1 score of 0.4784. It is important to note that BERT models, in general, tend to be computationally demanding, requiring both time and substantial GPU power for model testing. In [5], the authors proposed a methodology for linking CVEs to MITRE techniques through the utilization of two neural methods. The first method, LabelPowerset (neural), employed a Multilayer Perceptron as the base classifier, featuring two hidden layers and utilizing the softmax function for activation. The second method, BinaryRelevance (neural), also used a Multilayer Perceptron as the base classifier, incorporating two hidden layers and employing the sigmoid function for activation. Notably, the LabelPowerset (neural) model outperformed other classical models attempting the same task. This model's performance was evaluated using a dataset extracted from ENISA's threat report, comprising 27,471 vulnerabilities, with 8,077 linking CVEs to MITRE techniques and tactics. Out of those 8,077 links, 7,877 were used for training the model, and the remaining 200 were reserved for validation and analysis. Due to the dataset's limited size for training and testing, a 10-fold cross-validation method was employed to assess the model, followed by an analysis using the reserved 200 data points. The experiment demonstrated higher performance compared to other classical machine learning approaches, yielding an accuracy of 74.32%, which stands as the most interesting result in the related works. In [6], the authors employed a multi-headed joint-embedded neural model to address the task of linking CVEs to MITRE techniques, incorporating an unsupervised labeling technique. To embed CVE features into a latent space, three distinct embedding modules were utilized: a Token Encoder, a Label Encoder, and a Transformer-based Encoder. Samples from two domains, the feature domain of CVE and the corresponding ATT&CK domain, were projected into a joint latent space. This joint space captured the structure of labels, encoded texts, and interactions between the two. Subsequently, a Multi-layer Perceptron (MLP) classifier was applied to the joint latent space, providing independence from label set size. The resulting model possessed several noteworthy properties: (i) Each head of the model learned label dependencies from the perspectives of an attacker, defender, and CVE metadata. (ii) The joint latent space dimensions were made independent of label size, allowing the model to discover previously unseen labels crucial for the problem. (iii) The model was trained using the cross-entropy loss with a sigmoid function, well-suited for the challenges of a multi-label classification problem. In [7], the authors presented SMET that maps CVE entries (303) to ATT&CK techniques (41) based on their textual similarity. They achieved this by

[1]https://github.com/rocknrj/Linking-CVE-to-Mitre-Techniques

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leveraging ATT&CK BERT to learn semantic similarity. They utilized semantic extraction, ATT&CK BERT, and a logistic regression model to map CVE entries to ATT&CK techniques.

However, we could not include their model in our comparison since their model is not reproducible and the synthetic dataset used is not available. Furthermore, the metrics used to assess the quality of the trained model take into account partial mappings of the labels, in the sense that they considered the 5 best ranked labels to compute coverage error and Label Ranking Average Precision (LRAP).

In summary, a comparison of various models proposed in the literature was conducted. Table I provides an overview of the performance of each model, highlighting their respective accuracy and precision metrics, some remarks related to the solutions, the number of linked techniques and finally the reproducibility. Moreover, as previously mentioned, the CVET architecture, designed for linking CVEs to tactics and not techniques, was considered unsuitable for the present task. Models such as Multi-Label SciBERT by Grigorescu et al. [4] and LabelPowerset (neural) by Mendsaikhan et al. [5] align more closely with the objective at hand. Conversely, the Multi-Headed Joint Embedded Model was excluded as a benchmark method due to its use of unsupervised learning, requiring an additional labeling process (particularly challenging with multi-labeling), which is not the primary focus of our paper. Additionally, this model was only capable of linking to 17 techniques, falling short of the targeted objective of at least 50 techniques. However, as demonstrated in the experimentation section, our proposed solution, GuardLink, exhibited a slight performance improvement even when compared to this model.

III. Linking CVE ID to MITRE ATT&CK Architecture: GuardLink

Establishing a link between a CVE ID and a MITRE ATT&CK technique ID is a crucial undertaking in the realm of cybersecurity, to better understand and effectively address security threats.

A. Architecture

The general approach employed by GuardLink to accomplish this linkage is depicted in Fig. 1. Essentially, when a new CVE is introduced, the method utilizes Natural Language Processing (NLP) techniques to automatically extract the pertinent MITRE ATT&CK technique IDs from the MITRE ATT&CK matrix/knowledge base, enabling seamless association with the new CVE. The goal of GuardLink is to automatize this process and to provide comprehensive insights to the security teams to choose the best suited Mitigation techniques for every new input in the CVE database.

B. Motivating example

To better understand our research's motivation, hereinafter is a specific example. Let us consider the CVE-2020-7636 [8], which reports that the adb-driver (the Universal Android USB Driver) through 0.1.8 is vulnerable to Command Injection. The latter allows execution of arbitrary commands via the command function. In the MITRE ATT&CK framework, the technique ID "T1623" [9], which is associated with "Command and Scripting Interpreter", is more aligned with such a CVE. This technique is described as follows: "Adversaries may abuse command and script interpreters to execute commands, scripts, or binaries. These interfaces and languages provide ways of interacting with computer systems and are a common feature across many different platforms. Most systems come with some built-in command-line interface and scripting capabilities, for example, Android is a UNIX-like OS and includes a basic Unix Shell that can be accessed via the Android Debug Bridge (ADB) or Java's Runtime package". By linking CVE-2020-7636 to T1623, the security team will be able to enforce the appropriate mitigation techniques, namely: M1002 "Attestation" [10], which is described as follows: "Device attestation can often detect jailbroken or rooted devices" and M1010 "Deploy Compromised Device Detection Method" [11], which is described as follows: "Mobile security products can typically detect jailbroken or rooted devices". Furthermore, such a CVE can be detected using DS0041 "Application Vetting" [12] via API calls or via DS0009 "Process" [13] via monitoring of process creation and process metadata.

Finally, a CVE can be associated with one or more MITRE ATT&CK techniques. While in the previous example, only one technique was relevant (T1623), consider CVE-2023-5365 [14], which could be linked to multiple techniques, such as "T1626: Abuse Elevation Control Mechanism," "T1404: Exploitation for Privilege Escalation," and "T1631: Process Injection". This linkage enables cybersecurity experts to comprehensively analyze and respond to vulnerabilities, implementing appropriate mitigation, defenses, and detection measures tailored to known attack techniques. In practical terms, such linkages are instrumental in enhancing threat intelligence, incident response, and overall cybersecurity efforts.

IV. Data Processing

For our experiments, we mainly used the dataset provided by [5]. The dataset was taken from ENISA. It provided 27471 vulnerabilities out of which 8,077 links were found between CVEs and MITRE tactics and techniques. As for CVET model [4], the dataset used was from TagTog [15], which unfortunately was not accessible. But, we succeeded to retrieve some of the data uploaded in [16] for the BERT model [4]. This was used to test the model, however, the results vary as the obtained dataset might be missing some data.

A. Fixing the dataset

Once the dataset has been decided upon, the next step was to make sure that the linking of technique ID's with CVE and its description, is up to date with the current MITRE Framework version. The dataset from Mendsaikhan et al. [5] was collected in 2020 and since then MITRE framework has updated multiple times. Due to that, some technique IDs no longer exist while others were changed to new technique IDs. Therefore to further test the model on the latest framework, the dataset had to be fixed accordingly. Moreover, from the

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TABLE I
COMPARISON OF MODELS FROM LITERATURE BASED ON THEIR PERFORMANCES

Model	Accuracy	Precision	Main remarks	Nbr of links to techniques	Reproducibility
CVET [3]	76.93%	81.38	• Self-distillation knowledge approach • Pre-Trained Language Models: RoBERTa • Linked only to tactics (not techniques) • Unsupervised learning	0	No
Multi-Head Joint Embedded Model [6]	not provided	93.16%	• Unsupervised Learning / MLP • Deep learning / Classification	17	No
Augmented Multi-Label SciBERT [4]	not provided	59.26%	• Data Augmentation used • Supervised / multi-label DL classification / BERT	31	Yes
LabelPowerset (neural) [5]	74.32%	75.32%	• Deep learning / Classification • Multi-label classification / MLP	**52** (with 2 deprecated techniques)	No
GuardLink	**97.83%**	**97.63%**	• Deep learning / Classification • Multi-label classification / MLP	50	**Yes**

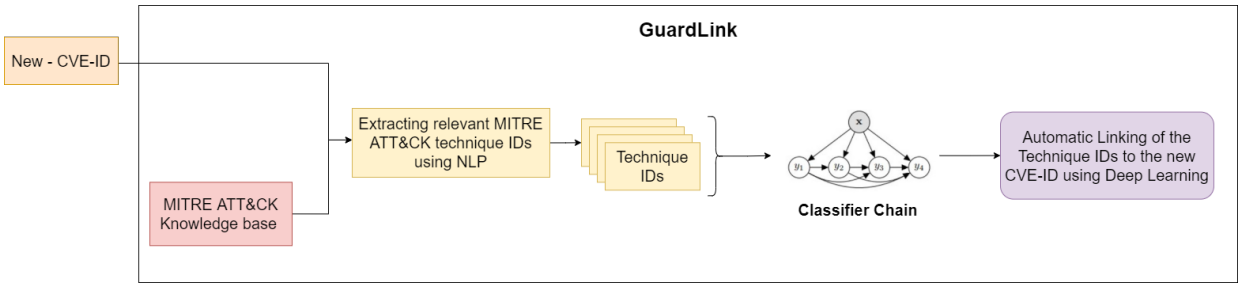


Fig. 1. GuardLink: Linking CVE to MITRE ATT&CK techniques general view

52 techniques that were linked to CVEs, 2 became deprecated and therefore there are only links to 50 techniques now. It also links to sub-techniques, as some of the techniques IDs that have been changed, links now to a sub-technique.

B. Data Augmentation

As the dataset was not large enough and was quite imbalanced, data augmentation was performed using the EasyDataAugmentor from the TextAttack module. Data augmentation is a process of increasing the dataset size by modifying the data. This form of text data augmentation changes the words of the CVE descriptions in the dataset and links the CVE ID as well as the technique ID to the augmented data. With this data augmentation and strategies, such as Iterative Stratification and StratifiedKFold to help with cross validation, the selected best new models along with the Labelpowerset (neural) model have been tested again to see how that impacted their performances.

C. Hyperparameter Tuning

This part involved changing the hyperparameters and testing how that impacted the performance of the model. This is done to see what parameters need to be set to achieve optimal results. It should also be noted that in the case of unknown/unshared hidden layer values, hyperparameter tuning is used to find a good hidden layer value.

V. EXPERIMENTING WITH NEW MODELS

The next step in our process consists in testing new classification models with the current dataset. The training and test sets were divided the same way as in [5]. We shortlisted three models: CatBoost, XGBoost and ClassifierChain.

Theses models were trained with 7,877 out of the 8,077 dataset links and the performance was tested on the 200 held out evaluation set and then compared with the best state of the art (denoted sota hereinafter) model (i.e., Labelpowerset (neural) [5]). To ensure a fair comparison, we employed the identical ratio as in [5]. Once a new model with good scores was found, a 10-fold cross validation was performed on it to further compare it with the best model.

A. Testing environment

The initial step involved establishing an environment conducive to replicating and testing the models presented in the preceding section, as well as experimenting with new models.

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For this testbed we used an HPC with a CPU 12th generation intel® core(tm) i9-12900k, 32 Gb RAM and a GPU Nvidia GP102 Titan XP. As for the software, Anaconda Navigator, Anaconda Command Line, and Jupyter Notebook were used for setting the environment and then testing the models. The core software used to implement the machine learning models was Python. Some of the main modules used are: scikit-learn, TensorFlow TextAttack, Matplotlib, tqdm, NumPy, Pandas and Beautiful Soup.

B. CatBoost

CatBoost [17] is a machine learning algorithm developed by Yandex. It is generally used for solving classification and regression problems. CatBoost is based on the gradient boosting algorithm, which is basically a collection of learning techniques that combines multiple weak learning, which are usually decision trees to create a strong predictive model. Gradient boosting builds trees sequentially, with each tree aiming to correct the errors of the previous tree.

However, as shown in Table II, the performance was below the best sota model and therefore did not provide any improvements. This model reached a lower accuracy of 56.93% on the evaluation set and improved the precision by 6.23 points of percentages.

TABLE II
CATBOOST PERFORMANCE ON EVALUATION SET

Model	Accuracy	Precision	Recall	F1 Score
Labelpowerset (neural)	**75.5%**	79.47%	**75.51%**	**77.44%**
CatBoost	56.93%	**85.70%**	64.11%	72.56%

C. XGBoost

XGBoost [18], which stands for Extreme Gradient Boosting, is also a machine learning algorithm that uses Gradient Boosting. Same as CatBoost, it is used for classification and regression tasks. It uses various techniques to deal with overfitting such as handling missing data using an adaptive algorithm, which learns how to treat missing values in a way that reduces its impact on the model's performance. However, unlike CatBoost, XGBoost requires one-hot encoding. However, there are extensions like "CatBoost encoding" that can be used to handle these categorical features more efficiently.

As shown in Table III, the performance seems to be better than CatBoost, it still is below the performance of best sota model. Therefore, the performance was deemed not to be good enough to further work on this model. This model scored a lower accuracy of 63% on the evaluation set, improved the precision by 10.33 points and the F1 by 1.18 point.

TABLE III
XGBOOST PERFORMANCE ON EVALUATION SET

Model	Accuracy	Precision	Recall	F1 Score
Labelpowerset (neural)	**75.5%**	79.47%	**75.51%**	77.44%
XGBoost	63%	**89.80%**	70.37%	**78.62%**

D. Classifier Chain

Classifier Chain [19] is a machine learning technique used to solve multi-label classification problems (i.e., when a data can be linked to one or more classes at the same time). The method addresses the multi-label classification by treating it as a series of binary classification problems. The Classifier Chain method integrates the computational efficiency of the binary relevance approach while maintaining the ability to consider label dependencies during the classification process.

Classifier Chain, first, creates a chain of binary classifiers where each classifier corresponds to one of the possible labels or classes. For example, if there are 2 labels A and B, it should create 2 binary classifiers: A vs. not A, and B vs. not B. Then the classifiers can be ordered according to one's needs. Each binary classifier is then trained to predict its corresponding label taking into account the features of the data and the labels predicted from the previous classifiers in the chain. This means that the prediction of a given label also depends on the predictions made for labels earlier in the chain after which the model can then be evaluated based on it's prediction performance. Count Vectorizer was used for feature extraction for this model.

The CVE descriptions were passed through the Count Vectorizer, while the technique ID's linked to them were passed through the MultiLabelBinarizer. This was then passed through Logistic Regression, which served as the base classifier for the model. Logistic Regression predicts the probability of an instance belonging to a specific class using a threshold value (commonly set at 0.5 by default) to convert these probabilities into binary predictions, which was then used as input for the Classifier Chain. From the 8,077 datasets, 7,877 was used to train the model and the remaining 200 were used to evaluate the performance. The obtained results are shown in Table IV. This model reached an accuracy of 84.5% on the evaluation set. For this reason, Classifier Chain serves as the primary predictive model for GuardLink.

Moreover, as mentioned earlier, since the dataset was updated to match the current version of the MITRE ATT&CK Framework [2] (the latter resulted in a loss of 2 techniques from the 54 linked MITRE techniques in the dataset), we ran other experiments on the Labelpowerset model (called "Labelpowerset (neural) Fixed"). Furthermore, in an attempt to improve the models, the hyperparameters of the two hidden layers were tuned to see if it could significantly improve the model (called "Labelpowerset (neural) Tuned"). The reproduced performance was found to be 2 points of percentage lesser than the actual results due to the unknown 200-item testing set, which was not shared and to the unknown values of the hidden layers. Finally, we applied data augmentation on the dataset to further improve the models performances, called "Augmented Labelpowerset (neural)".

Table IV shows a significant increase in accuracy compared to the best sota model, Labelpowerset (neural). To further attempt to compare the performance between these two models, 10-fold cross validation was attempted on the Classifier Chain.

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TABLE IV
CLASSIFIER CHAIN VS LABELPOWERSET (NEURAL) PERFORMANCE ON EVALUATION SET

Model	Accuracy	Precision	Recall	F1 Score
Labelpowerset (neural)	75.5%	79.47%	75.51%	77.44%
Augmented Labelpowerset (neural)	95.37%	95.92%	94.69%	95.30%
Classifier Chain	84.5%	84.72%	86%	85.35%
Augmented Classifier Chain	**97.83%**	**97.63%**	**97.37%**	**97.51%**

However, due to the class imbalance in the dataset, 1 of the 10 folds failed to find 2 classes to perform testing. Hence the accuracy of one fold could not be measured.

This drove us to search for a technique to fix the class imbalance to properly measure the 10 fold cross validation. Furthermore, the hyperparameters of the Classifier Chain and Logistic regression were also tested with the following details being specified (and the rest left as default):

- Classifier Chain = order = random, random_state=None
- Logistic Regression = solver = newton-cg/liblinear, random_state=None

Both solvers for logistic regression (libLinear and newton-cg) performed equally in terms of accuracy. However, due to the fact that our linking process is considered complex and non linear, and as the dataset improves, newton-cg is the preferred solver for this classification task. Therefore, we used the 5 cross validation methods to compare our model with the Labelpowerset one, namely: cross_validate, cross_val_predict, KFold (k=10), StratifiedKFold and MultilabelStratifiedKFold as shown in Table VI, Table V, Table VII, Table VIII and Table IX respectively.

The findings demonstrate that the Classifier Chain model outperforms the Labelpowerset (neural) model across all performance metrics. This is shown through its predictions on the evaluation set and when comparing the performance results obtained through various cross-validation methods. Additionally, employing stratification with Data Augmentation has led to a slight improvement in addressing class imbalance, resulting in a more equitable scoring system during cross-validation.

VI. CONCLUSION AND FUTURE WORKS

We proposed, in this paper, GuardLink, a solution based on Classifier Chain, to further improve the linking of CVE IDs to MITRE ATT&CK framework [2], more specifically MITRE techniques IDs. Among the state of the art proposals, we compared our solution with the Labelpowerset (neural) [5], which showed the best performance for the classification tasks for such a linking. The dataset for this model was taken from ENISA's "State of Vulnerabilities 2018/2019" report, which contains 27,471 CVE data out of which 8,077 were linked to MITRE tactics and techniques. This was the main dataset used for our benchmark. As shown in our experiments, Classifier Chain was found to be the best performing model, outperforming the Labelpowerset (neural) model.

In this research, the most challenging aspect in creating an accurate model was the quality of the datatset. The dataset encapsulated only 50 out of the 201 techniques. Therefore, improving the balance of the dataset, as well as the quality and the number of links in the dataset is among our future work.

REPRODUCIBILITY

Ensuring the reproducibility of scientific research is paramount in fostering transparency and trust within the academic and technical communities. As a testament to our commitment to this principle, we have made our code and datasets accessible on our GitHub repository [20].

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TABLE V
CROSS_VAL_PREDICT 10-FOLD CROSS VALIDATION COMPARISON

Model (CV = cross_validate)	Accuracy	Micro Average			Macro Average			Hamming Loss	Ranking Loss
		Precision	Recall	F1 Score	Precision	Recall	F1 Score		
Labelpowerset (neural) Original	0.7432	0.7532	0.7380	0.7452	0.6827	0.6264	0.6396	0.0911	0.2448
Labelpowerset (neural) Tuned	0.7335	0.7382	0.7358	0.7367	0.6168	0.5995	0.6003	N/A	N/A
Labelpowerset (neural) Fixed	0.7325	0.7390	0.7291	0.7338	0.6144	0.5812	0.5902	N/A	N/A
Augmented Labelpowerset (neural) Fixed	0.7218	0.7272	0.7250	0.7259	0.6060	0.5772	0.5821	N/A	N/A
Augmented Classifier Chain	**0.9786**	**0.9781**	**0.9753**	**0.9767**	**0.9451**	**0.9390**	**0.9419**	N/A	N/A

TABLE VI
CROSS_VALIDATE 10-FOLD CROSS VALIDATION COMPARISON

Model (CV = cross_val_predict)	Accuracy	Micro Average			Macro Average			Hamming Loss	Ranking Loss
		Precision	Recall	F1 Score	Precision	Recall	F1 Score		
Labelpowerset (neural) Paper	0.7432	0.7532	0.7380	0.7452	0.6827	0.6264	0.6396	0.0911	0.2448
Labelpowerset (neural) Tuned	0.7183	0.7398	0.7464	0.7431	0.6516	0.6095	0.6235	0.0854	0.2537
Labelpowerset (neural) Fixed	0.7287	0.7335	0.7444	0.7389	0.6297	0.5980	0.6062	0.0857	0.2612
Augmented Labelpowerset (neural) Fixed	0.7158	0.7254	0.7324	0.7289	0.6194	0.5903	0.6008	0.0888	0.2667
Augmented Classifier Chain	**0.9787**	**0.9769**	**0.9737**	**0.9753**	**0.9827**	**0.9751**	**0.9789**	**0.0080**	**0.0212**

TABLE VII
KFOLD CROSS VALIDATION COMPARISON

Model (CV = KFold)	Micro Average			Macro Average			Hamming Loss	Ranking Loss
	Precision	Recall	F1 Score	Precision	Recall	F1 Score		
Labelpowerset (neural) Paper	0.7532	0.7380	0.7452	0.6827	0.6264	0.6396	0.0911	0.2448
Augmented Classifier Chain	**0.9774**	**0.9748**	**0.9761**	**0.9832**	**0.9771**	**0.9801**	**0.0077**	**0.0206**

TABLE VIII
STRATIFIEDKFOLD 10-FOLD CROSS VALIDATION COMPARISON

Model (CV = StratifiedKFold)	Accuracy	Micro Average			Macro Average			Hamming Loss	Ranking Loss
		Precision	Recall	F1 Score	Precision	Recall	F1 Score		
Labelpowerset(neural) Paper	0.7432	0.7532	0.7380	0.7452	0.6827	0.6264	0.6396	0.0911	0.2448
Labelpowerset(neural) Tuned	0.7682	0.7839	0.7733	0.7784	0.6876	0.6534	0.6610	0.0727	0.2230
Labelpowerset(neural) Fixed	0.7648	0.7814	0.7685	0.7746	0.6726	0.6419	0.6498	0.0727	0.2272
Augmented Labelpowerset(neural) Fixed	0.9621	0.9610	0.9629	0.9619	0.9266	0.9244	0.9250	0.0124	0.0363
Augmented Classifier Chain	**0.9781**	**0.9783**	**0.9735**	**0.9759**	**0.9461**	**0.9378**	**0.9419**	**0.0078**	**0.0213**

TABLE IX
MULTILABELSTRATIFIEDKFOLD 10-FOLD CROSS VALIDATION COMPARISON

Model (CV = MultilabelStratifiedKFold	Accuracy	Micro Average			Macro Average			Hamming Loss	Ranking Loss
		Precision	Recall	F1 Score	Precision	Recall	F1 Score		
Labelpowerset(neural) Paper	0.7432	0.7532	0.7380	0.7452	0.6827	0.6264	0.6396	0.0911	0.2448
Labelpowerset(neural) Tuned	0.7737	0.7902	0.7710	0.7805	0.7134	0.6531	0.6737	0.0716	0.2176
Labelpowerset(neural) Fixed	0.7675	0.7729	0.7791	0.7758	0.6853	0.6551	0.6626	0.0733	0.2235
Augmented Labelpowerset(neural) Fixed	0.9630	0.9634	0.9624	0.9629	0.9295	0.9211	0.9248	0.0120	0.0356
Augmented Classifier Chain	**0.9797**	**0.9784**	**0.9772**	**0.9778**	**0.9514**	**0.9459**	**0.9486**	**0.0072**	**0.0195**

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